

Dawei (David) Liu

Research focus: Reliable tool-using AI systems: skill retrieval, execution assurance, and multimodal workflows.

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EDUCATION

University of Pennsylvania

M.S.E. in Computer & Information Science | **GPA: 3.90/4.00**

Aug 2024 - May 2026

Philadelphia, PA

Northeastern University

B.E. in Software Engineering | **Rank: 5/396** (Sep 2023)

Sep 2020 - Jun 2024

Shenyang, China

RESEARCH & OPEN-SOURCE EXPERIENCE

Graph-of-Skills (GoS)

EMNLP 2026

First author; proposed the dependency-graph formulation, built the public retrieval system, and led evaluation.

Combined hybrid semantic-lexical seeding, reverse-weighted Personalized PageRank, and budget-aware loading.

Achieved **25.6% higher reward** (+7.0 absolute points) and **56.7% fewer total tokens** vs. full skill loading on 1,000-skill SkillsBench (GPT-5.2 Codex).

Dr. Claw

EMNLP 2026 Demo

Core contributor and active maintainer; built the cross-platform **Electron desktop application** and release pipeline for an open-source AI scientist workspace. Implemented provider integrations, project/session workflows, and reliability fixes.

COMFYCLAW

ECCV 2026 MUCG

Co-first author; built the core agent loop for typed edits to ComfyUI workflow graphs, the integration plugin, and experiment framework. Ran two model evaluations and wrote the first draft.

PROFESSIONAL EXPERIENCE

ByteDance Inc.

Jun 2026 - Present

Software Engineer, Commerce Ads

Seattle, WA

- **Agent operations:** Built a production on-call agent that triages Lark alert threads using SOPs, runtime telemetry, change history, and service-topology context.
- **Agent interoperability:** Built an inference gateway with Responses/Messages API compatibility, connecting Claude Code/Codex-compatible clients to an internal model platform while preserving sandboxing, approval policies, and multi-agent workflows.
- **Agent validation:** Built a resumable validation workflow with commit-traceable runtime and artifact checks, validation gates, side-effect deduplication, and execution audit trails.

ByteDance Inc.

May 2025 - Dec 2025

Software Engineer Intern, Commerce Ads

Seattle, WA

- Developed image-selection ranking for Shop Ads using posterior features, exploration-exploitation, and multimodal LLM quality evaluation.
- Automated creative generation for high-value products via scheduled workflows and Flink-backed asset persistence; built modular preview/diagnostics with test-entity injection for ad-format and rollout validation.
- Improved serving reliability and tail latency through caching, asynchronous batch fetches, and offline feature computation.

Amazon

Jun 2024 - Aug 2024

Software Development Engineer Intern, Global-Mile Team

Beijing, China

- Built a ByteBuddy-based OpenTelemetry Java agent to instrument distributed requests without application changes, capturing payloads and traces.
- Correlated traces by business ID across partially instrumented services; dynamically selected Kinesis or SQS transport for Fargate and Lambda runtimes.
- Built a React/Java diagnostics console with searchable traces, timelines, and payload views.

JD Technology

Jul 2023 - Oct 2023

Software Engineer Intern, Algorithm Tools

Beijing, China

- Redesigned Kubernetes resource management with ConfigMaps, Informers, asynchronous updates, and sharded locks, reducing cold-start overhead.
- Modularized campaign-page generation with the Strategy pattern and generative AI; improved multi-tenant delivery with Argo Workflows, Argo CD, and Helm.

SELECTED PUBLICATIONS

- Graph-of-Skills: Dependency-Aware Structural Retrieval for Massive Agent Skills** 2026
Dawei Liu[†], Zongxia Li[†], Hongyang Du, Xiyang Wu, Shihang Gui, Yongbei Kuang, Lichao Sun.
EMNLP 2026 (Main Conference); AgentSkills 2026 Poster, ACM CAIS. arXiv:2604.05333. [paper] [code]
Highest average reward among evaluated methods in all six SkillsBench/ALFWorld model-benchmark blocks.
- COMFYCLAW: Self-Evolving Skill Harnesses for Image Generation Workflows** 2026
Zongxia Li[†], Dawei Liu^{*}, Fuxiao Liu, Yuhang Zhou, Xiyang Wu, et al.
2nd MUCG Workshop at ECCV 2026. arXiv:2607.01709. [paper] [code]
- Dr. Claw: An AI Scientist Workspace for Vibe Research** 2026
Dingjie Song, Hanrong Zhang, Dawei Liu, Yixin Liu, Zongxia Li, Zhengqing Yuan, et al.
EMNLP 2026 (System Demonstrations). arXiv:2609.00365. [paper] [code]
- A Cookbook of 3D Vision: Data, Learning Paradigms, and Application** 2026
Hongyang Du^{*}, Zongxia Li[†], Dawei Liu^{*}, Runhao Li^{*}, Haoyuan Song, Qingyu Zhang, et al.
CVPR 2026 Workshop OpenSUN3D. pp. 8376-8390. [paper] [code]
- Multimodal Video Generation Models with Audio: Present and Future** 2026
Dawei Liu, Hongyang Du, Zongxia Li, Xiyang Wu, Lantao Yu, Jingxi Chen, et al.
ACM Multimedia 2026 Workshop JAV-CG. [paper] [code]
- TIMEDB: tumor immune micro-environment cell composition database with automatic analysis and interactive visualization** 2023
Xueying Wang^{*}, Lingxi Chen^{*}, Wei Liu^{*}, Yuanzheng Zhang, Dawei Liu, et al.
Nucleic Acids Research. 51(D1):D1417-D1424. [paper] [site]
Developed the visualization framework and several visualization analyses. [team contribution]

^{*} Equal contribution; [†] Core contribution.

RESEARCH IMPACT & SERVICE

- Citations.** 88 citations across six selected works (Sep 14, 2026).
- Research use.** GoS: 14 papers use its public assets, reimplement its methods, compare against it, or test its components (Sep 12, 2026).
- Open-source interest.** GoS: 208 stars, 26 forks; Dr. Claw: 1,105 stars, 123 forks (Sep 14, 2026).
- Peer review.** Reviewer, ACM CAIS Workshop AgentSkills (2026); reviewed two papers.

SELECTED HONORS

- Winner, Apple Swift Student Challenge** 2026
Built **Aura Chef**, a touchless Swift playground integrating hand-pose recognition and voice commands; selected as one of 350 winners from 37 countries and regions.
- China Collegiate Computing Contest - Mobile Application Innovation** 2022
Team lead & sole software developer of *Trace Note* (iOS/AR); National Final Second Prize and Northeast Division Innovation Track First Prize.
- National Scholarship for Undergraduate Students** 2020-2021
Ministry of Education, China.

TECHNICAL SKILLS

Languages	Python, Go, Java, C/C++, JavaScript/TypeScript, SQL, Swift
Agent/ML	LLM agents, tool calling, retrieval/GraphRAG, multimodal evaluation, PyTorch, ranking/recommendation
Systems/Infra	Distributed services, OpenTelemetry, Kafka, Flink, MySQL, Redis, Elasticsearch, Thrift/Protobuf
Cloud/DevOps	Linux, Docker, Kubernetes, AWS, Helm, CI/CD, Grafana, Git